CUSTOMER CHURN PREDICTION USING MACHINE LEARNING

Submitted by

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in partial fulfillment of the requirements for the award of the degree

of

MASTER'S OF BUSINESS ADMINISTRATION

In

UNITED UNIVERSITY

विद्या परमं बलम्

Session 2026-27

UNDER THE GUIDANCE OF

Mr. Ferhan Ahmed

United University

1. TITLE

Project Title

Customer Churn Prediction Using Machine Learning

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Course / Programme: MBA - IBM
University / College: United University
Submitted to: IBM
Academic Session: 2025-2027

Project Overview

This project studies how machine learning can help a telecommunications business identify customers who may discontinue its services. The report follows the ten headings prescribed for the assignment: Title, Introduction, Problem Statement, Discuss Data, Method, Tools & Technologies Used, Workflow, Procedure, Predictive Model, Insights, and Recommendation.

2. INTRODUCTION

2.1 Introduction to Customer Churn

Customer churn refers to a situation in which a customer stops using a company's product or service. In telecommunications, churn may occur when a customer cancels a phone or internet subscription, moves to another provider, or does not renew a service. Churn is important because companies invest time and money in acquiring customers, supporting them, and maintaining service relationships. When customers leave, the company may lose recurring revenue and may need to spend additional money to attract replacement customers.

2.2 Meaning of Predictive Analytics

Predictive analytics uses historical data, statistical techniques, and machine learning algorithms to estimate an outcome that is not yet known. It does not guarantee what will happen. Instead, it produces a probability or class based on patterns learned from previous examples. In this project, the outcome is whether a customer is labelled as having churned or not churned in the dataset.

2.3 Role of Machine Learning

Machine learning allows a computer system to learn relationships between customer characteristics and known churn outcomes. A model may use variables such as contract type, customer tenure, monthly charges, payment method, and subscribed services. After training, the model can estimate churn risk for customers whose outcome is not supplied. Business teams can use these estimates to prioritize customer-support calls, review pricing concerns, or offer suitable service options.

2.4 Business Importance

Churn analysis supports customer relationship management, service quality improvement, and better use of retention budgets. A company may not be able to contact every customer personally. A ranked list of higher-risk customers can help the company decide where to investigate first. However, predictions should be combined with customer feedback and business judgement; a high-risk score is not proof that a customer intends to leave.

2.5 Aim and Objectives

The main aim is to develop a clear educational workflow for predicting customer churn using a public telecommunications dataset. The objectives are to understand the churn problem; describe the dataset and its variables; prepare data for analysis; explore patterns in customer behaviour; explain a suitable classification model; define appropriate evaluation measures; interpret results responsibly; and recommend actions that may improve customer retention.

2.6 Scope of the Project

The project focuses on a public, sample telecommunications dataset and on binary classification: Churn = Yes or Churn = No. It is suitable for a classroom project and for demonstrating the predictive analytics lifecycle. It is not a production-ready system. The dataset is a snapshot rather than a sequence of monthly observations, so it does not by itself establish a reliable forecast for a particular future date. The model also cannot prove that a retention offer will cause a customer to stay.

3. PROBLEM STATEMENT

3.1 Business Problem

Telecommunications providers operate in competitive markets where customers can compare plans, prices, service quality, and contract terms. Some customers discontinue their service, and the business may not know which customers are at greater risk until after cancellation. A general retention campaign may waste money by giving discounts to customers who would have stayed anyway, while failing to help customers who genuinely need support.

3.2 Problem Definition

The problem is to use available customer information to classify each customer into one of two groups: likely churn / churn-labelled customer, or likely to remain / non-churn-labelled customer. The target variable in the selected dataset is named Churn and contains Yes and No values. The model learns from labelled examples and estimates the class for held-out records.

3.3 Why the Problem Matters

Retaining customers can protect recurring revenue and preserve customer relationships. Identifying patterns associated with churn may also reveal service or billing issues that need attention. For example, a business might investigate whether customers with short tenure or flexible monthly contracts show different churn rates from long-standing customers with longer contracts. Such patterns are associations, not necessarily causes.

3.4 Research Questions

The project is designed to address these questions: (1) What information is available about each customer? (2) How is churn distributed in the dataset? (3) Which customer and account characteristics are associated with churn? (4) How can a classification model be trained and evaluated? (5) Which metrics are useful when false predictions have different business costs? (6) How can a company use predictions responsibly to improve retention?

3.5 Project Objectives

Specific objectives are to examine the structure and quality of the data; convert fields into suitable data types; encode categorical variables; separate training and testing data; train a baseline classification algorithm; evaluate it with a confusion matrix and multiple metrics; review influential patterns; and develop practical recommendations.

3.6 Assumptions and Limitations

The project assumes that the dataset fields are accurately recorded and that the observed relationships are useful for a classroom demonstration. It is limited by the dataset’s age, context, and snapshot design. The records describe a fictional telecommunications business in California, so results should not be generalized automatically to every company, country, or customer population. A model should be retrained and validated using recent data before real-world use.

4. DISCUSS DATA

4.1 Dataset Source

This report uses the IBM Telco Customer Churn sample dataset, which is publicly available through Kaggle and other data repositories. The commonly used version contains 7,043 customer records and 21 columns, including a customer identifier, customer attributes, service details, account and billing information, and the Churn target. Source link: https://www.kaggle.com/datasets/blastchar/telco-customer-churn. Dataset information and schema are also documented in public technical references.

4.2 Unit of Analysis and Data Structure

Each row represents one customer. Each column represents an attribute describing that customer or the observed churn outcome. The file is commonly supplied as a CSV (comma-separated values) file. A CSV can be opened in spreadsheet software or imported into Python, R, or analytics software. The customerID column identifies records but should normally be excluded from model training because an identifier does not provide a meaningful behavioural signal by itself.

4.3 Target Variable

The target variable is Churn. The value Yes indicates that the customer left within the period represented by the sample; No indicates that the customer did not leave in that period. This is a binary classification task. During modelling, the target can be encoded as 1 for Yes and 0 for No, while the original labels should be retained for understandable reporting.

4.4 Main Variables

Demographic and household variables include gender, SeniorCitizen, Partner, and Dependents. Tenure records how long the customer has been with the company, in months. Service variables include PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, and StreamingMovies. Account and billing variables include Contract, PaperlessBilling, PaymentMethod, MonthlyCharges, and TotalCharges.

50 Customer Records

Read Me

customerID	gender	tenure_months	Contract	InternetService	PaymentMethod	MonthlyCharges	TotalCharges	Churn
7590-VHVEG	Female	1	Month-to-month	DSL	Electronic check	29.85	29.85	Yes
5575-GNVDE	Male	34	One year	DSL	Mailed check	56.95	1889.50	No
3668-QPYBK	Male	2	Month-to-month	Fiber optic	Electronic check	53.85	108.15	Yes
7795-CFOCW	Male	45	One year	DSL	Bank transfer (automatic)	42.30	1840.75	No
9237-HQITU	Female	2	Month-to-month	Fiber optic	Electronic check	70.70	151.65	Yes
9305-CDSKC	Female	8	Month-to-month	Fiber optic	Electronic check	99.65	820.50	Yes
1452-KIOVK	Male	22	Month-to-month	Fiber optic	Credit card (automatic)	89.10	1949.40	No
6713-OKOMC	Female	10	Month-to-month	DSL	Mailed check	29.75	301.90	No
7892-POOKP	Female	28	Month-to-month	Fiber optic	Electronic check	104.80	3046.05	Yes
6388-TABGU	Male	62	One year	DSL	Bank transfer (automatic)	56.15	3487.95	No
9763-GRSKD	Male	13	Month-to-month	DSL	Mailed check	49.95	587.45	No
7469-LKBCI	Male	16	Two year	No	Credit card (automatic)	18.95	326.80	No
8091-TTVAX	Male	58	One year	Fiber optic	Credit card (automatic)	100.35	5681.10	No
0280-XJGEX	Male	49	Month-to-month	Fiber optic	Bank transfer (automatic)	103.70	5036.30	Yes
5129-JLPIS	Male	25	Month-to-month	Fiber optic	Electronic check	105.50	2686.05	No
3655-SNQYZ	Female	69	Two year	Fiber optic	Credit card (automatic)	113.25	7895.15	No
8191-XWSZG	Female	52	One year	No	Mailed check	20.65	1022.95	No
9959-WOFKT	Male	71	Two year	Fiber optic	Bank transfer (automatic)	106.70	7382.25	No
8717-ERXIC	Female	1	Month-to-month	Fiber optic	Electronic check	99.60	99.60	Yes
9928-JPRMC	Female	21	Month-to-month	Fiber optic	Electronic check	89.10	1862.90	No
4190-MFLUW	Female	12	One year	DSL	Credit card (automatic)	69.40	853.10	No
4183-MYFRB	Female	5	Month-to-month	DSL	Electronic check	69.40	348.80	Yes
8779-QRDMV	Male	1	Month-to-month	Fiber optic	Electronic check	94.70	94.70	Yes
1680-VDCWW	Male	49	Two year	No	Bank transfer (automatic)	20.15	973.35	No
CUST-0025	Male	7	Month-to-month	Fiber optic	Electronic check	85.40	597.80	Yes
CUST-0026	Female	41	One year	DSL	Bank transfer (automatic)	54.20	2222.20	No
CUST-0027	Male	19	Month-to-month	No	Mailed check	20.10	381.90	No
CUST-0028	Female	3	Month-to-month	Fiber optic	Electronic check	95.60	286.80	Yes
CUST-0029	Male	64	Two year	DSL	Credit card (automatic)	61.30	3923.20	No
CUST-0030	Female	11	Month-to-month	Fiber optic	Electronic check	102.50	1127.50	Yes
CUST-0031	Male	36	One year	DSL	Bank transfer (automatic)	48.75	1755.00	No
CUST-0032	Female	2	Month-to-month	No	Mailed check	20.35	40.70	No
CUST-0033	Male	55	Two year	Fiber optic	Credit card (automatic)	108.25	5953.75	No
CUST-0034	Female	9	Month-to-month	DSL	Electronic check	67.80	610.20	Yes
CUST-0035	Male	27	One year	Fiber optic	Bank transfer (automatic)	91.40	2467.80	No
CUST-0036	Female	1	Month-to-month	DSL	Electronic check	49.25	49.25	Yes
CUST-0037	Male	46	Two year	No	Credit card (automatic)	19.80	910.80	No
CUST-0038	Female	15	Month-to-month	Fiber optic	Electronic check	98.75	1481.25	Yes
CUST-0039	Male	33	One year	DSL	Mailed check	57.60	1900.80	No
CUST-0040	Female	6	Month-to-month	Fiber optic	Electronic check	101.20	607.20	Yes
CUST-0041	Male	70	Two year	Fiber optic	Bank transfer (automatic)	110.30	7721.00	No
CUST-0042	Female	24	One year	DSL	Credit card (automatic)	63.50	1524.00	No
CUST-0043	Male	4	Month-to-month	Fiber optic	Electronic check	88.90	355.60	Yes
CUST-0044	Female	59	Two year	No	Bank transfer (automatic)	20.25	1194.75	No
CUST-0045	Male	17	Month-to-month	DSL	Mailed check	50.80	863.60	No
CUST-0046	Female	8	Month-to-month	Fiber optic	Electronic check	97.30	778.40	Yes
CUST-0047	Male	38	One year	Fiber optic	Credit card (automatic)	92.60	3518.80	No
CUST-0048	Female	14	Month-to-month	DSL	Electronic check	68.40	957.60	No
CUST-0049	Male	2	Month-to-month	Fiber optic	Electronic check	99.10	198.20	Yes
CUST-0050	Female	66	Two year	DSL	Bank transfer (automatic)	59.75	3943.50	No

4.5 Data Dictionary

customerID: unique customer identifier; gender: recorded gender category; SeniorCitizen: indicator for senior-citizen status; Partner: whether the customer has a partner; Dependents: whether the customer has dependents; tenure: number of months with the company; PhoneService: whether phone service is active; MultipleLines: multiple-line service status; InternetService: internet service category; OnlineSecurity, OnlineBackup, DeviceProtection and TechSupport: optional service status; StreamingTV and StreamingMovies: streaming service status; Contract: contract length category; PaperlessBilling: paperless billing status; PaymentMethod: payment method; MonthlyCharges: monthly billed amount; TotalCharges: total charges recorded; Churn: target outcome.

4.6 Data Types

The dataset contains both numerical and categorical variables. Numerical variables include tenure, MonthlyCharges, and TotalCharges. Binary categorical fields often contain Yes/No values, while fields such as Contract, InternetService, and PaymentMethod contain multiple categories. SeniorCitizen is represented as a numeric indicator. Correctly identifying data types is necessary because many algorithms require categorical fields to be encoded into numerical form.

4.7 Data Quality Considerations

In the common version, TotalCharges may be stored as text rather than as a numeric field. A small number of records have blank TotalCharges values, often associated with customers whose tenure is zero. During preprocessing, convert TotalCharges to numeric and inspect the resulting missing values. Depending on the project design, these values can be set to zero when justified by the data meaning, imputed with a documented method, or the affected records can be excluded. The chosen method should be recorded.

4.8 Class Distribution

The commonly used dataset contains 1,869 churn-labelled customers and 5,174 non-churn customers. This is approximately 26.5% churn and 73.5% non-churn. The classes are therefore not equally represented. A model that predicts every customer as non-churn would achieve about 73.5% accuracy but would fail to identify churners. This illustrates why accuracy alone is insufficient and why recall, precision, F1-score, and the confusion matrix should also be reviewed.

4.9 Example Data Rows

A small dataset preview can be inserted here after downloading the CSV. Use a few anonymized or sample rows and show only relevant columns such as tenure, Contract, MonthlyCharges, TotalCharges, and Churn. Do not include unnecessary personal information. The preview should be labelled as a sample of the dataset, not as a result produced by the model.

4.10 Data Preparation Plan

The preparation plan is to import the file; inspect row and column counts; check duplicates; review missing and invalid values; convert TotalCharges to a numeric type; identify categorical and numerical features; exclude customerID from the predictors; encode categorical variables; split the data into training and test subsets using stratification; and document every transformation. The transformations should be fitted on training data only to avoid data leakage.

5. METHOD, TOOLS & TECHNOLOGIES USED

5.1 Analytical Approach

The project combines descriptive analysis and predictive analytics. Descriptive analysis summarizes the dataset through counts, percentages, charts, and comparisons. Predictive analysis trains a classification model to estimate the churn label. The descriptive stage helps the student understand the data before building the model, while the predictive stage evaluates whether patterns learned from training examples generalize to unseen test records.

5.2 Software and Tools

Recommended tools include Python with pandas for data handling, NumPy for numerical operations, Matplotlib or Seaborn for visualizations, and scikit-learn for model training and evaluation. IBM SPSS Modeler can also be used to build a visual workflow with data import, type checking, partitioning, model nodes, and evaluation nodes. Excel may be used for initial inspection and simple summaries. Choose the tool actually used and update this section accordingly.

5.3 Data Cleaning

Data cleaning includes checking missing values, inconsistent categories, duplicate identifiers, incorrect data types, and impossible values. TotalCharges should be converted from text to numeric. Blank values should be handled using a documented rule. The target column should be checked for unexpected labels. Cleaning should be performed consistently and without using the test set to make decisions that would not be available in a real prediction scenario.

5.4 Encoding Categorical Variables

Most machine learning algorithms require numeric inputs. One-hot encoding can convert a categorical field such as Contract into separate indicator columns for the available categories. Binary fields can be encoded as 0 and 1. A consistent encoder should be used for both training and test data so that the same categories map to the same columns.

5.5 Train-Test Split

A typical educational setup uses 80% of the records for training and 20% for testing. Stratified splitting maintains approximately the same churn proportion in both subsets. The model learns from the training set, while the test set is reserved for final evaluation. A fixed random seed can make results reproducible. If extensive tuning is required, use cross-validation on the training data and retain the test set for final assessment.

5.6 Candidate Algorithms

Logistic Regression is a useful baseline for binary classification and can provide interpretable coefficients after suitable preprocessing. A Decision Tree creates rule-like splits and is relatively easy to explain, although it can overfit. Random Forest combines many decision trees and can model non-linear patterns, but may be less transparent than a single tree. The project can begin with Logistic Regression or a Decision Tree and compare a second model if time permits.

5.7 Evaluation Metrics

Accuracy is the proportion of all predictions that are correct. Precision measures how many customers predicted as churners actually churned. Recall measures how many actual churners were identified. The F1-score combines precision and recall. ROC-AUC measures how well the model ranks positive cases across thresholds. A confusion matrix shows true positives, false positives, true negatives, and false negatives. The best threshold depends on the business cost of missed churners and unnecessary retention offers.

5.8 Ethical and Responsible Use

Customer churn scores should support, not replace, human judgement. Businesses should protect customer data, restrict access, explain the purpose of analysis, and avoid unfair treatment. Demographic variables should be considered carefully and checked for unfair differences in model errors. A high-risk score should not be used to penalize a customer or deny a service. Retention offers should be tested for effectiveness and fairness.

6. WORKFLOW

6.1 Overall Workflow

The project follows a structured process: define the business problem → obtain the dataset → inspect the data → clean and transform fields → conduct exploratory data analysis → select features → split training and test data → train a baseline model → generate predictions → evaluate performance → interpret findings → recommend retention actions.

6.2 Workflow Explanation

Step 1: Define the churn problem and the intended use of the model. Step 2: Download the public dataset and record its source. Step 3: Inspect the number of records, columns, data types, and target values. Step 4: Clean missing or invalid values. Step 5: Explore churn rates across relevant customer groups. Step 6: Prepare features through encoding and appropriate transformations. Step 7: Split the data into training and test sets. Step 8: Train a classifier. Step 9: Predict labels or probabilities for test records. Step 10: Evaluate the model with multiple metrics. Step 11: Interpret the strongest patterns. Step 12: formulate practical, testable recommendations.

6.3 Suggested Flowchart

Start → Load dataset → Validate data → Clean data → Explore patterns → Prepare features → Split data → Train model → Predict churn → Evaluate metrics → Explain insights → Recommend retention actions → End.

6.4 Inputs and Outputs

Inputs include customer tenure, contract type, service subscriptions, billing information, payment method, and the historical churn label used during training. Outputs include a predicted class and, where supported, a churn probability. The output should be accompanied by model evaluation results and a clear note that predictions are uncertain. A business user should receive understandable summaries rather than only raw model scores.

7. PROCEDURE

7.1 Step 1: Obtain the Dataset

Download the Telco Customer Churn CSV from the dataset link. Store the original file unchanged and create a working copy for analysis. Record the source, access date, file name, and version. This improves reproducibility and helps other students understand which data was used.

7.2 Step 2: Import and Inspect

Import the CSV into the selected tool. Check the number of rows and columns, field names, data types, and sample records. Confirm that Churn contains the expected Yes and No values. Check whether customerID is unique. Summarize numerical variables using minimum, maximum, mean, median, and standard deviation where useful.

7.3 Step 3: Clean the Data

Convert TotalCharges to numeric values and inspect blanks created by conversion. Review whether blanks correspond to customers with zero tenure. Apply the selected missing-value rule and record it. Check duplicates and category spelling. Avoid deleting records automatically without considering why the values are missing.

7.4 Step 4: Conduct Exploratory Data Analysis

Calculate the overall churn rate. Compare churn proportions by Contract, tenure groups, InternetService, TechSupport, OnlineSecurity, and PaymentMethod. Create a bar chart of churn versus non-churn, a bar chart of churn rate by contract type, and a histogram of tenure. These summaries help reveal patterns that can guide modelling and business interpretation.

7.5 Step 5: Prepare Features

Set Churn as the target variable and remove customerID from model inputs. Separate numerical and categorical predictors. Convert categorical predictors into model-ready numerical columns using one-hot encoding or the selected software's equivalent. If numerical scaling is used, estimate scaling parameters from the training data only.

7.6 Step 6: Split Training and Test Data

Split the dataset into training and test subsets, for example 80% training and 20% testing. Use stratification on Churn to preserve class proportions. Keep the test set separate until model selection is complete. Document the split ratio and random seed.

7.7 Step 7: Train a Baseline Model

Train a Logistic Regression or Decision Tree classifier on the training subset. If using Logistic Regression, use a preprocessing pipeline that encodes categorical features and scales numerical features when appropriate. If using a tree-based method, scaling is usually not required. Set a reproducible random seed for algorithms that use randomness.

7.8 Step 8: Generate Predictions

Use the trained model to predict churn labels for the test set. If the model provides probabilities, save the predicted probability of Churn = Yes. These probabilities can be used to rank customers by estimated risk. A probability is not a guarantee and should not be treated as a customer's definite intention.

7.9 Step 9: Evaluate the Model

Create a confusion matrix and calculate accuracy, precision, recall, F1-score, and ROC-AUC where probability scores are available. Review false negatives carefully because they represent churners the model failed to identify. Review false positives because they may trigger unnecessary retention contacts or discounts. Select a threshold based on business objectives rather than accuracy alone.

7.10 Step 10: Interpret and Document

Summarize which customer groups have higher or lower observed churn rates. Compare these findings with model behaviour. If feature importance or coefficients are used, explain that they show associations within the fitted model and do not establish causality. Record limitations, recommended actions, and what additional data would improve the model.

7.11 Example Python Procedure

The following is a compact implementation outline. It is provided as a reproducible starting point; it should be run against the downloaded CSV before reporting any actual model metrics.

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, confusion_matrix, roc_auc_score

# Load data
df = pd.read_csv('WA_Fn-UseC_-Telco-Customer-Churn.csv')
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
df = df.drop(columns=['customerID'])
y = df.pop('Churn').map({'No': 0, 'Yes': 1})
X = df

num_cols = X.select_dtypes(include=['number']).columns
cat_cols = X.select_dtypes(exclude=['number']).columns
numeric_pipe = Pipeline([('imputer', SimpleImputer(strategy='median')), ('scale', StandardScaler())])
categorical_pipe = Pipeline([('imputer', SimpleImputer(strategy='most_frequent')), ('encode',
OneHotEncoder(handle_unknown='ignore'))])
preprocess = ColumnTransformer([('num', numeric_pipe, num_cols), ('cat', categorical_pipe, cat_cols)])
model = Pipeline([('prep', preprocess), ('classifier', LogisticRegression(max_iter=1000,
class_weight='balanced'))])
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, stratify=y, random_state=42)
model.fit(X_train, y_train)
pred = model.predict(X_test)
prob = model.predict_proba(X_test)[:, 1]
print(confusion_matrix(y_test, pred))
print(classification_report(y_test, pred))
print('ROC-AUC:', roc_auc_score(y_test, prob))
```

7.12 Software Procedure in IBM SPSS Modeler

If IBM SPSS Modeler is used, import the CSV through a source node, inspect field roles and measurement levels with a Type node, and clean or derive fields using appropriate preparation nodes. Use a partition node to separate training and testing records. Connect the training stream to a Logistic Regression or C5.0/Decision Tree modelling node, run the model, and evaluate predictions using an analysis or evaluation node. The exact node names and available options may vary by version. Save screenshots of the stream, model settings, and output for the final report.

8. PREDICTIVE MODEL

8.1 Model Design

The predictive task is binary classification. The model receives customer attributes as input and estimates the probability that the target label is Churn = Yes. A Logistic Regression model is a suitable baseline because it is widely used for binary outcomes and can produce probabilities. A Decision Tree can be included as a second model because its decision rules are easier to visualize. Model choice should be guided by held-out performance, interpretability, and the cost of different mistakes.

8.2 Logistic Regression

Logistic Regression estimates the log-odds of the positive class as a weighted combination of input features. A sigmoid function converts this value into a probability between zero and one. A threshold, commonly 0.5 as a starting point, converts the probability into a class label. The threshold can be adjusted when the business values recall more than precision, or vice versa. Coefficients should be interpreted only after considering feature encoding, scaling, and the other variables in the model.

8.3 Decision Tree

A Decision Tree splits records into groups based on feature values. It may produce rules involving contract type, tenure, or service status. Trees are easy to explain but can overfit if they grow too deep. Maximum depth, minimum leaf size, and pruning can help control complexity. A tree should be evaluated on unseen data rather than judged by its training accuracy.

8.4 Model Training

During training, the algorithm uses input features and known target labels to learn patterns. The training set is not a final measure of performance because the model has already seen those records. If hyperparameters are selected, use cross-validation within the training data. The test set should be used once for the final unbiased assessment after decisions are finalized.

8.5 Confusion Matrix

A confusion matrix compares actual and predicted classes. True Positive (TP): a churner correctly identified. True Negative (TN): a non-churner correctly identified. False Positive (FP): a customer predicted to churn who did not churn in the labelled outcome. False Negative (FN): a churner incorrectly predicted to stay. For a retention programme, false negatives can be costly because a customer at risk may not receive attention; false positives can also waste resources.

8.6 Performance Metrics

Accuracy = (TP + TN) / (TP + TN + FP + FN). Precision = TP / (TP + FP). Recall = TP / (TP + FN). F1-score = 2 × (Precision × Recall) / (Precision + Recall). ROC-AUC summarizes ranking performance across thresholds. Report the confusion matrix and several metrics together. If classes are imbalanced, accuracy alone can make a weak model look stronger than it is.

8.7 Results Reporting Template

After running the model, enter the actual test-set results below. Do not leave illustrative figures in place as if they were measured results.

Metric | Actual value
Accuracy | _____
Precision for Churn = Yes | _____
Recall for Churn = Yes | _____
F1-score for Churn = Yes | _____
ROC-AUC | _____
True Positives | _____
False Positives | _____
True Negatives | _____
False Negatives | _____

8.8 Illustrative Example Only

The following small example explains how a confusion matrix is interpreted; it is not a measured result from this project. Suppose a test set contains 200 customers and the model identifies 30 actual churners correctly, misses 10 churners, correctly classifies 140 non-churners, and incorrectly flags 20 non-churners. Then TP = 30, FN = 10, TN = 140, and FP = 20. Accuracy would be (30 + 140) / 200 = 85%; precision would be 30 / (30 + 20) = 60%; recall would be 30 / (30 + 10) = 75%. These values are purely illustrative and must not be reported as the project's actual performance.

8.9 Model Limitations

The dataset is a cross-sectional snapshot and does not include a long sequence of dated customer observations. The model therefore predicts the provided churn label from available attributes; it does not prove what will happen next month. Results may change with different train-test splits, preprocessing choices, thresholds, or algorithms. Changes in prices, competitors, service quality, and customer behaviour can make a model less accurate over time.

8.10 Model Monitoring

If a company deployed a model, it should monitor performance over time, check whether churn rates and customer behaviour have changed, review fairness, and retrain when necessary. The business should also measure whether retention interventions actually reduce churn. A model can rank customers at risk without identifying the best action for each customer; treatment effectiveness requires additional experimentation.

9. INSIGHTS

9.1 Overall Churn Distribution

The commonly used dataset contains 1,869 churn-labelled customers out of 7,043, or approximately 26.5%. The remaining 5,174 customers, approximately 73.5%, are labelled as non-churn. This class imbalance means that a useful model must do more than predict the majority class. Recall and precision for churn should be examined alongside accuracy.

9.2 Contract Type

Prior analyses of the Telco Customer Churn dataset commonly observe higher churn among month-to-month customers than among customers with one-year or two-year contracts. This pattern is plausible because monthly contracts provide more flexibility to leave. For this report, calculate the churn rate within each contract group from the downloaded file and insert a chart. Treat the pattern as an association, not proof that contract length alone causes churn.

9.3 Customer Tenure

Tenure indicates how long a customer has stayed with the provider. Churn often differs between newer and longer-standing customers. A tenure histogram and churn-rate-by-tenure-group chart can help show whether the earliest customer relationship period deserves additional attention. A company could examine onboarding quality, setup difficulties, and early support requests for newer customers.

9.4 Billing and Payment

MonthlyCharges, TotalCharges, and PaymentMethod may help describe customer account patterns. Higher charges should not automatically be interpreted as dissatisfaction; the amount may reflect a more comprehensive service package. Compare churn rates across billing groups and payment methods, and consider service mix when interpreting differences.

9.5 Additional Services

Fields such as OnlineSecurity, OnlineBackup, DeviceProtection, and TechSupport allow the analyst to compare customers with and without additional services. Some groups may show different churn rates. These differences can help generate questions about perceived value, service quality, and customer needs. They should not be treated as proof that adding a service will prevent churn.

9.6 Interpreting Model Features

If the fitted model provides coefficients or feature importance, list the leading features and describe their direction or relative contribution where interpretation is valid. One-hot encoded categories must be interpreted with their reference categories and other model variables in mind. Feature importance can be unstable when variables are correlated. Model explanations should be checked against business knowledge.

9.7 Operational Insight

A useful operational output is a ranked list of customers by predicted churn probability, combined with relevant service and account information. The list can help customer service teams prioritize outreach. The company should define a clear threshold and outreach capacity rather than assuming that every customer above a threshold should receive the same offer.

9.8 Evidence to Add

For a complete submission, add actual outputs from the analysis: (1) churn distribution chart, (2) churn rate by contract type, (3) tenure distribution or tenure-group comparison, (4) confusion matrix, and (5) actual evaluation metrics. Every chart should have a descriptive title, labelled axes, and a brief interpretation. Do not present general observations as if they were computed from your own run unless you have verified them.

10. RECOMMENDATION

10.1 Retention Strategy for High-Risk Customers

Use the model to prioritize customer outreach, not to make automatic decisions about customers. A support representative can ask whether the customer has experienced service problems, billing confusion, or difficulty using the product. The response should guide the next step. High-risk customers should receive helpful service rather than pressure or an assumption that they will leave.

10.2 Improve the Early Customer Experience

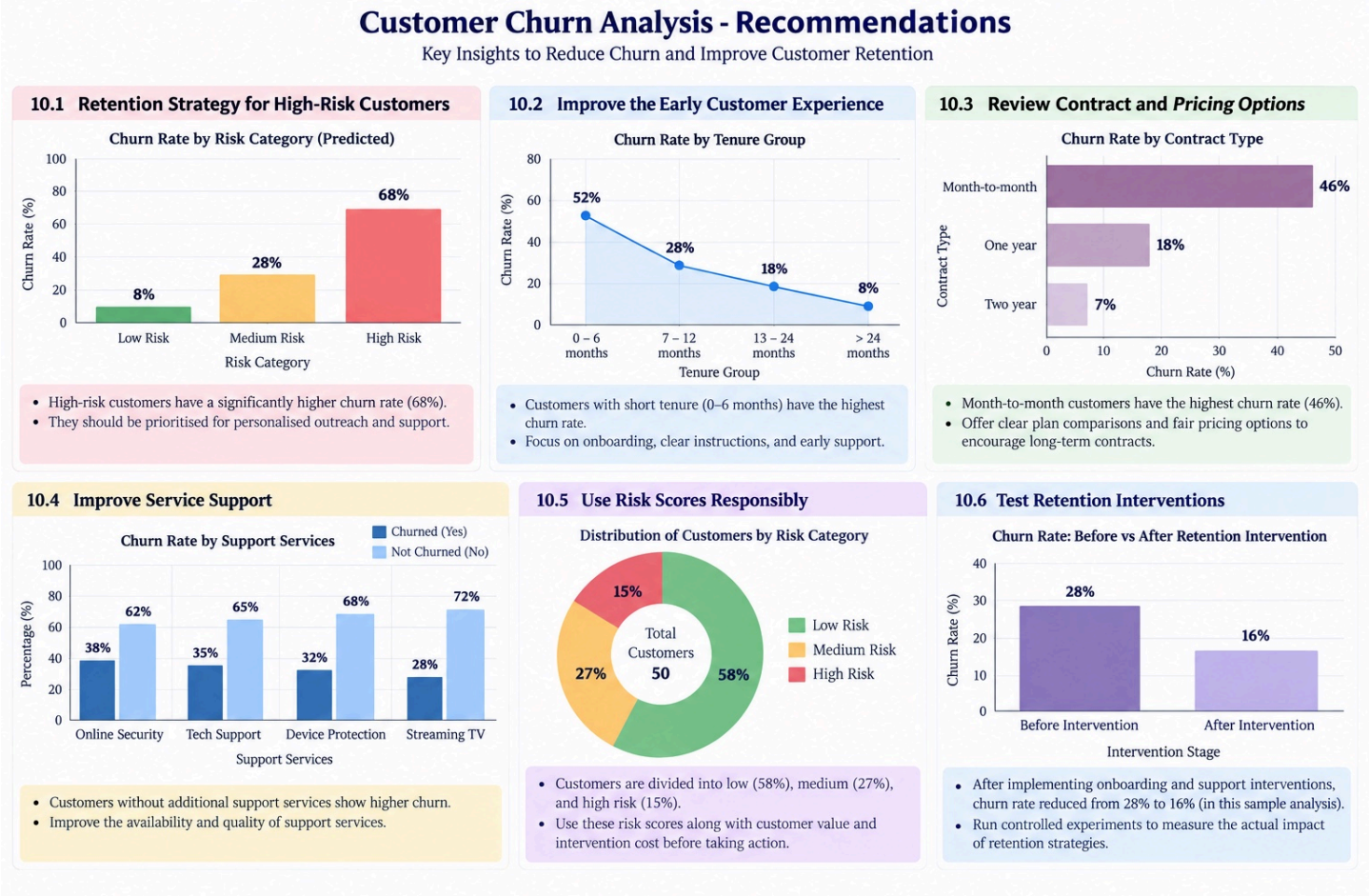
If the analysis shows elevated churn among customers with short tenure, strengthen onboarding. Provide clear installation instructions, a welcome message, transparent billing information, and an accessible route to technical support. Follow up after activation to identify unresolved issues. The effectiveness of these steps should be measured rather than assumed.

10.3 Review Contract and Pricing Options

If month-to-month customers show higher churn, review whether they are dissatisfied with price, service quality, or flexibility. Offer clear plan comparisons and fair contract options. Avoid forcing customers into longer contracts; explain the benefits and restrictions transparently. Retention offers should be targeted according to customer needs and tested for business value.

10.4 Improve Service Support

If customers without technical support or security services show higher churn, investigate whether the difference reflects unmet needs, product complexity, or customer segment differences. Improve the availability and quality of support. Explain optional services in plain language and recommend them only when relevant to the customer.



10.5 Use Risk Scores Responsibly

Set practical risk categories such as low, medium, and high only after examining the distribution and validating thresholds. Combine risk with customer value and the cost of intervention. A high probability of churn does not mean a customer should automatically receive a discount. Sometimes resolving a service issue is more effective and less expensive than reducing the price.

10.6 Test Retention Interventions

Run controlled experiments where feasible. For example, compare an eligible group receiving a new onboarding or support intervention with a comparable group receiving the standard process. Measure churn, customer satisfaction, cost per retained customer, and long-term value. This helps determine whether the intervention actually changes outcomes rather than merely targeting customers who were already likely to stay.

10.7 Data Privacy and Fairness

Limit access to customer records, use data only for a legitimate purpose, and avoid sharing identifiable customer information unnecessarily. Check whether model errors differ across relevant groups. Document the model's intended use, limitations, and review process. Provide a human review path for important customer decisions.

10.8 Future Scope

Future work can compare Logistic Regression, Decision Tree, Random Forest, and gradient-boosting models; tune thresholds based on intervention costs; use more recent customer data; include complaint history and service outages if legally and ethically appropriate; and monitor model drift. A future study could also estimate customer lifetime value and test which retention action works best for different customer segments.

10.9 Conclusion

Customer churn prediction is a useful application of predictive analytics because it converts historical customer information into estimates of churn risk. The project demonstrates how to define a business problem, inspect and prepare data, train a classification model, evaluate its performance, interpret patterns, and suggest practical retention actions. The most important lesson is that model performance must be measured carefully and used responsibly. A prediction is a decision-support signal, not a guarantee or a causal explanation.

10.10 References

1. Kaggle, Telco Customer Churn dataset: https://www.kaggle.com/datasets/blastchar/telco-customer-churn
2. IBM sample data repository reference: https://github.com/IBM/telco-customer-churn-on-icp4d
3. Scikit-learn documentation, model evaluation: https://scikit-learn.org/stable/modules/model_evaluation.html
4. Scikit-learn documentation, Logistic Regression: https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
5. Scikit-learn documentation, preprocessing data: https://scikit-learn.org/stable/modules/preprocessing.html
6. Dataset schema and class distribution reference: https://github.com/cat-cols/telco-churn-analytics/blob/main/docs/data_dictionary.md

Accessed for educational report preparation: October 2026.